

Javascript Array

```
var num = [1, 2, 3, 4];
num.at(1)           // 2
num.push(5)         // Add element to the end: [1, 2, 3, 4, 5]
num.pop()           // remove last element: [1, 2, 3]
num.fill(1)         // fill every element: [1, 1, 1, 1]
num.shift()         // remove first element: [2, 3, 4]
num.unshift(5)      // Add element to beginning: [5, 1, 2, 3, 4 ]
num.reverse()       // sort in descending order: [4, 3, 2, 1]
num.includes(2)     // is array contains a specified value: true
num.map( item => 2*item) // Map elements; [2, 4, 6, 8]
num.filter( item => item > 2) // filter element: [3, 4]
num.find(item => item > 2) // Find element : 3(first match)
num.every(item => item > 0) // true
num.findIndex(item => item === 2) // 1
num.reduce( (prev, curr) => prev + curr, 0) // 10
num.toString();     // convert to string
num.join(" * ");    // join: "1 * 2 * 3 * 4"
num.splice(2, 0, "i", "p"); // add elements : [1, 2, 'i', 'p', 3, 4]
num.slice(1,4);      // slice elements from [1] to [4-1]
num.sort();          // sort string alphabetically
x.sort(function(a, b){return a - b}); // numeric sort
x.sort(function(a, b){return b - a}); // numeric descending sort
x.sort(function(a, b){return 0.5 - Math.random()}); // random sort
```

Miscellaneous

- setTimeout()
- setInterval()
- clearTimeout()
- clearInterval()
- parseInt()
- parseFloat()
- JSON.stringify()
- JSON.parse()
- typeof
- instanceof

Function Types

- Regular Functions
- Arrow Functions
- Anonymous Functions
- Immediately Invoked Function Expressions (IIFE)
- Callback Functions
- Higher-Order Functions

Number Methods

- `.toFixed()`
- `.toPrecision()`
- `.toString()`
- `Number.isInteger()`
- `Number.isNaN()`

Event Methods

- `.addEventListener()`
- `.removeEventListener()`
- `.stopPropagation()`
- `.preventDefault()`

Promise Methods

- `Promise.all()`
- `Promise.race()`
- `Promise.resolve()`
- `Promise.reject()`
- `.then()`
- `.catch()`
- `.finally()`

Error Handling

- `try`
- `catch`
- `finally`
- `throw`

- Math.abs()
- Math.ceil()
- Math.floor()
- Math.round()
- Math.max()
- Math.min()
- Math.random()
- Math.pow()
- Math.sqrt()
- Math.trunc()

Date Methods

- .getDate()
- .getDay()
- .getFullYear()
- .getHours()
- .getMinutes()
- .getSeconds()
- .getMilliseconds()
- .getMonth()
- .getTime()
- .setDate()
- .setFullYear()
- .setHours()
- .setMinutes()
- .setSeconds()
- .setMilliseconds()
- .setMonth()
- .setTime()

Array Methods

- .push()
- .pop()
- .shift()
- .unshift()
- .concat()
- .slice()
- .splice()
- .indexOf()
- .lastIndexOf()
- .includes()
- .find()
- .findIndex()
- .filter()
- .map()
- .reduce()
- .forEach()
- .some()
- .every()
- .sort()
- .reverse()
- .join()

Object Methods

- Object.keys()
- Object.values()
- Object.entries()
- Object.assign()
- Object.freeze()
- Object.seal()
- Object.create()
- Object.hasOwnProperty()
- Object.getPrototypeOf()

Data Types

- String
- Number
- Boolean
- Array
- Object
- Null
- Undefined
- Symbol
- BigInt

Control Flow

- if
- else
- else if
- switch
- break
- continue

Loops

- for
- while
- do...while
- for...of
- for...in

String Methods

- .charAt()
- .concat()
- .includes()
- .indexOf()
- .lastIndexOf()
- .slice()
- .split()
- .substring()
- .toLowerCase()
- .toUpperCase()
- .trim()
- .replace()
- .startsWith()
- .endsWith()

Javascript Events

Media

```
onabort, oncanplay, oncanplaythrough,  
ondurationchange, onended, onerror,  
onloadeddata, onloadedmetadata,  
onloadstart, onpause, onplay, onplaying,  
onprogress, onratechange, onseeked,  
onseeking, onstalled, onsuspend,  
ontimeupdate, onvolumechange, onwaiting
```

Miscellaneous

```
transitionend, onmessage, onmousewheel,  
ononline, onoffline, onpopstate, onshow,  
onstorage, ontoggle, onwheel,  
ontouchcancel, ontouchend, ontouchmove,  
ontouchstart
```

Javascript Events

Mouse

```
onclick, oncontextmenu, ondblclick,  
onmousedown, onmouseenter, onmouseleave,  
onmousemove, onmouseover, onmouseout,  
onmouseup
```

Keyboard

```
onkeydown, onkeypress, onkeyup
```

Frame

```
onabort, onbeforeunload, onerror,  
onhashchange, onload, onpageshow,  
onpagehide, onresize, onscroll, onunload
```

Form

```
onblur, onchange, onfocus, onfocusin,  
onfocusout, oninput, oninvalid, onreset,  
onsearch, onselect, onsubmit
```

Drag

```
ondrag, ondragend, ondragenter,  
ondragleave, ondragover, ondragstart,  
ondrop
```

Animation

```
animationend, animationiteration,  
animationstart
```

Javascript JSON

```
var str = '{"names":[" + // crate JSON object
  '{"first":"Hakuna","lastN":"Matata" },' +
  '{"first":"Jane","lastN":"Doe" },' +
  '{"first":"Air","last":"Jordan" }]}';

myObj = JSON.parse(str); // parse
document.write(myObj.names[1].first); // access
var myJSON = JSON.stringify(myObj); // stringify
ocalStorage.setItem("testJSON", myJSON); // storing data
text = localStorage.getItem("testJSON"); // retrieving data
```

Javascript Object

```
var student = { // object name
  firstName:"Jane", // list of properties and values
  lastName:"Doe",
  age:18,
  height:170,
  fullName : function() { // object function
    return this.firstName + " " + this.lastName;
  }
};
student.age = 19; // setting value
student[age]++; // incrementing
name = student.fullName(); // call object function
```

Errors Handlings

```
try {           // block of code to try
    undefinedFunction();
}
catch(err) {    // block to handle errors
    console.log(err.message);
}
```

```
// Input validation

var x = document.getElementById("mynum").value;

try {
    if(x == "") throw "empty";           // error cases
    if(isNaN(x)) throw "not a number";
    x = Number(x);
    if(x > 10) throw "too high";
}
catch(err) {                             // if there's an error
    document.write("Input is " + err);    // output error
    console.error(err);                   // write the error in console
}
finally {
    // executed regardless of the try / catch result
    document.write("</br />Done");
}
```

Javascript Dates

```
var d = new Date();

Date("2017-06-23");           // date declaration
Date("2017");                 // is set to Jan 01
Date("2017-06-23T12:00:00-09:45"); // YYYY-MM-DDTHH:MM:SSZ
Date("June 23 2017");         // long date format
Date("Jun 23 2017 07:45:00 GMT+0530"); // time zone

a = d.getDay();              // getting the weekday

getDate();                   // day as a number (1-31)
getDay();                    // weekday as a number (0-6)
getFullYear();               // four digit year (yyyy)
getHours();                  // hour (0-23)
getMilliseconds();           // milliseconds (0-999)
getMinutes();                // minutes (0-59)
getMonth();                  // month (0-11)
getSeconds();                // seconds (0-59)
getTime();                   // milliseconds since 1970
```